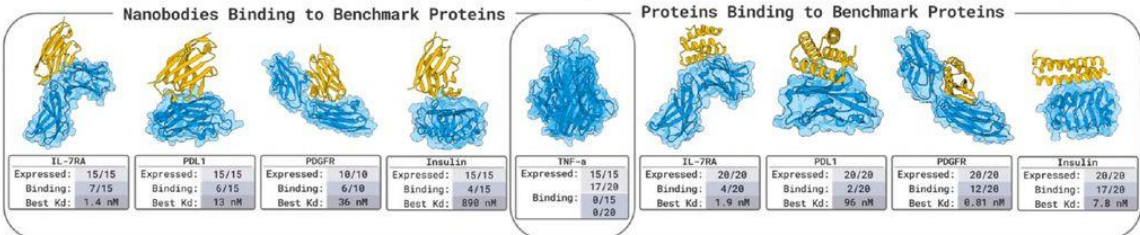
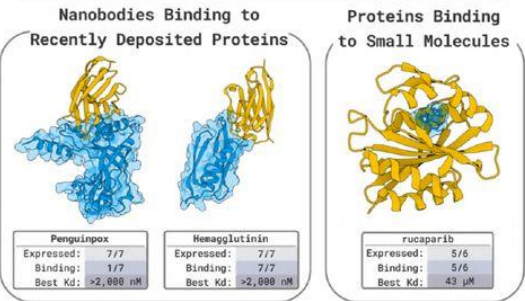
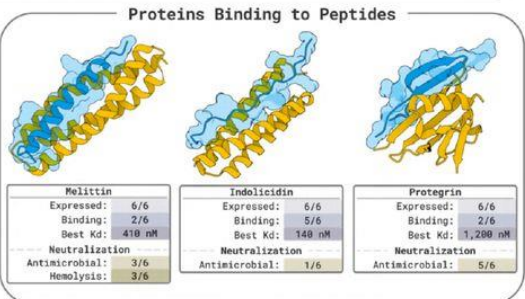
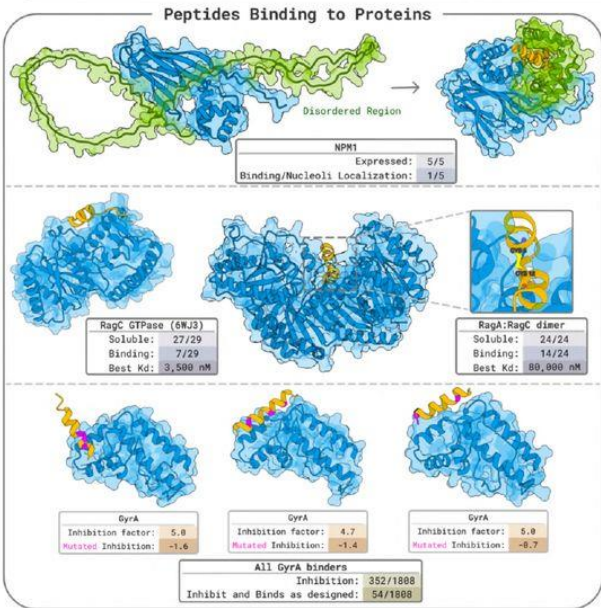
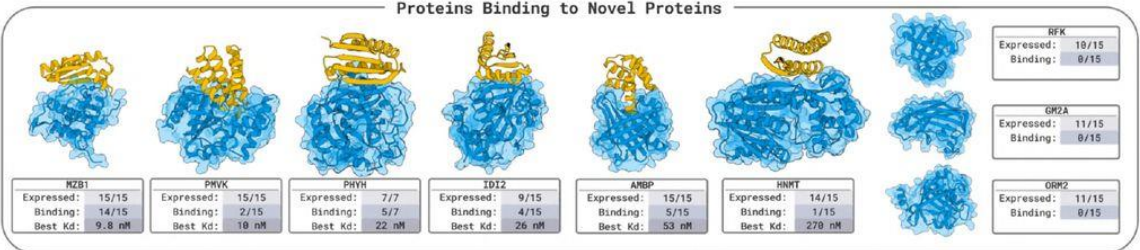
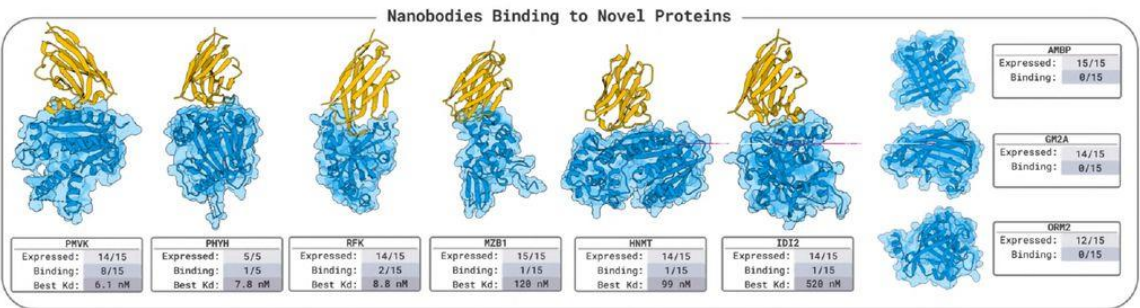


Workshop 07 – Part 2

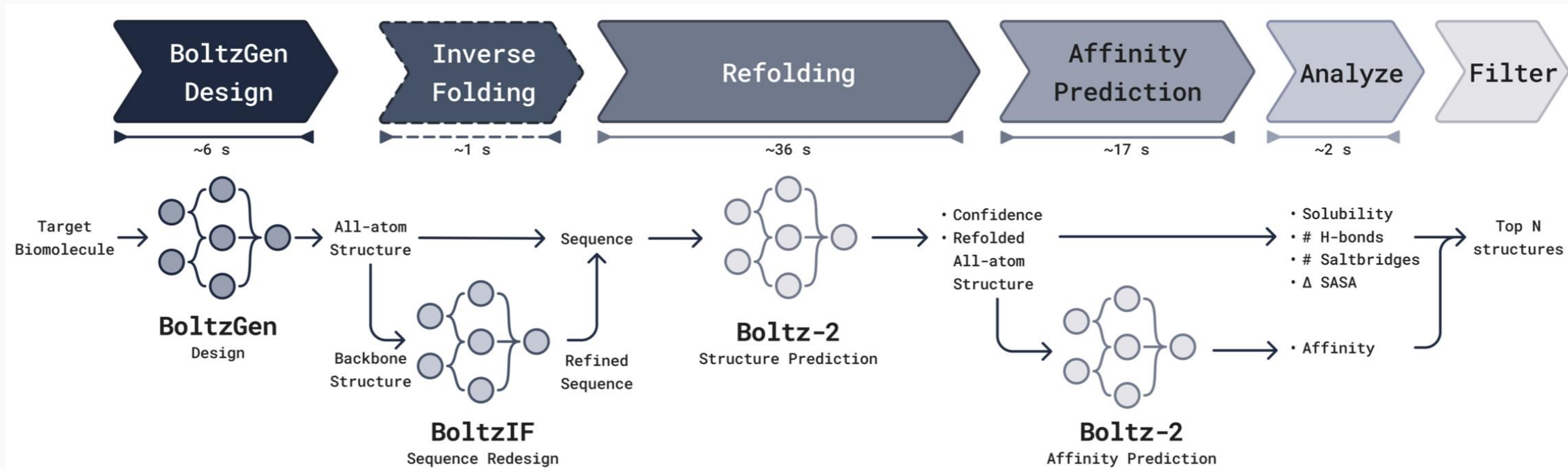
Protein Design using Boltzgen



Boltzgen



Boltzgen



RFDiffusion
pipeline

RFDiffusion

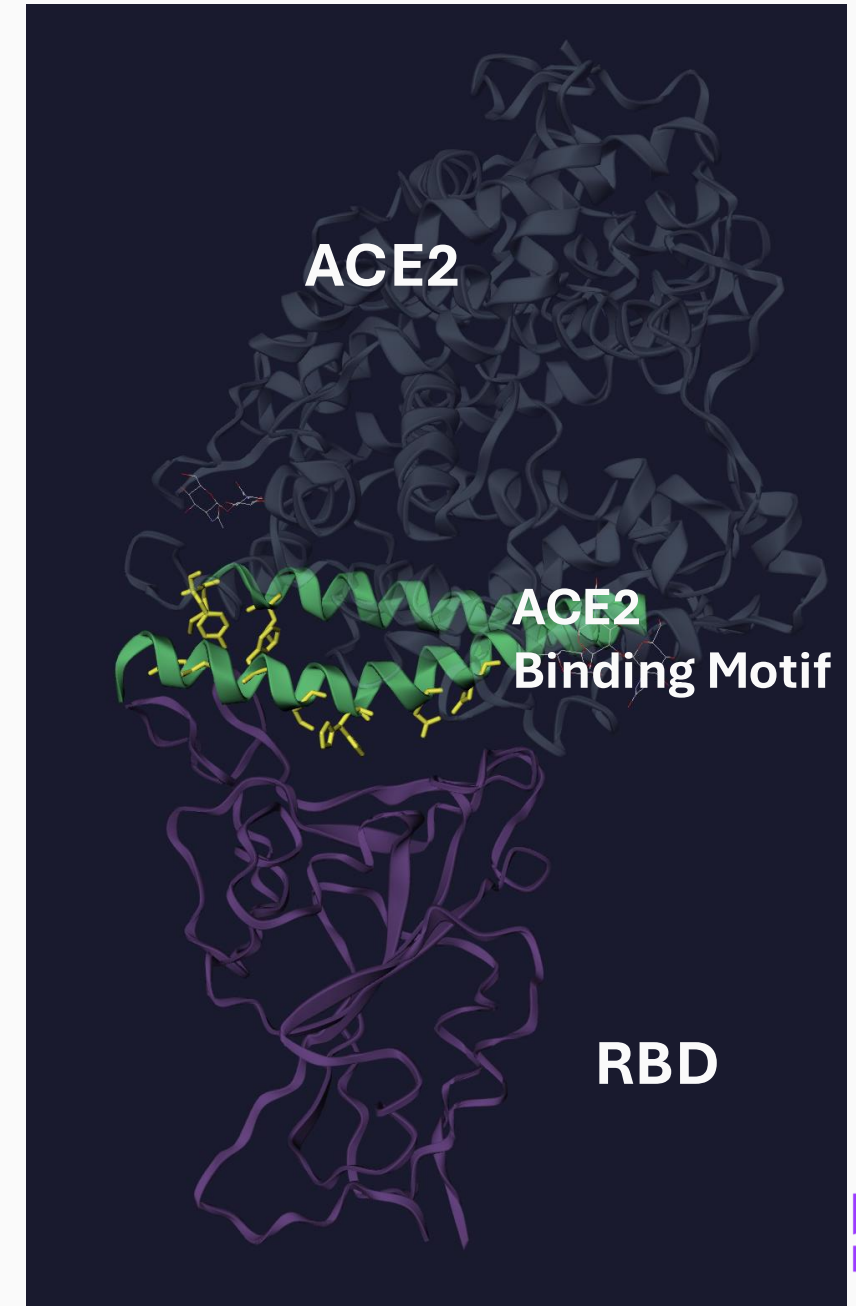
ProteinMPNN



Design Aim

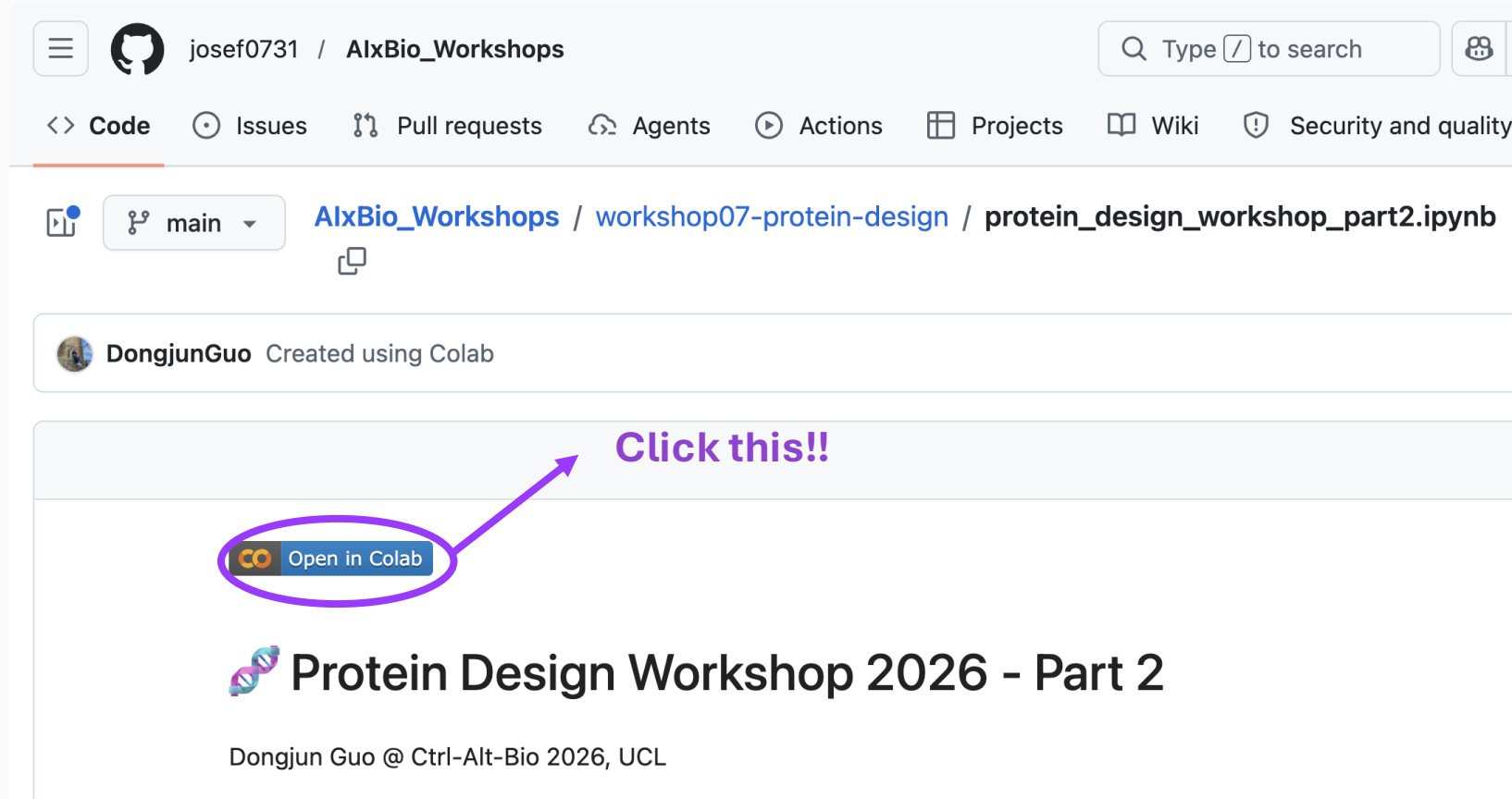
Design a binder for the RBD using ACE2
Binding Motif as the template

- Re-design the two helices except the yellow hotspots
- Design an extended third helix at the C terminal



Colab demonstration

Open the google colab: go to github, under workshop07 folder, you will see a file `protein\_design\_workshop\_part2.ipynb`



The screenshot shows the GitHub interface for the repository 'josef0731 / AlxBio_Workshops'. The file 'protein_design_workshop_part2.ipynb' is selected. A purple arrow points to the 'Open in Colab' button, which is circled in purple. The text 'Click this!!' is written in purple next to the arrow. Below the button, the title 'Protein Design Workshop 2026 - Part 2' is displayed, followed by the subtitle 'Dongjun Guo @ Ctrl-Alt-Bio 2026, UCL'.

GitHub repository: josef0731 / AlxBio_Workshops

File path: AlxBio_Workshops / workshop07-protein-design / protein_design_workshop_part2.ipynb

Created using Colab by DongjunGuo

Click this!!

[Open in Colab](#)

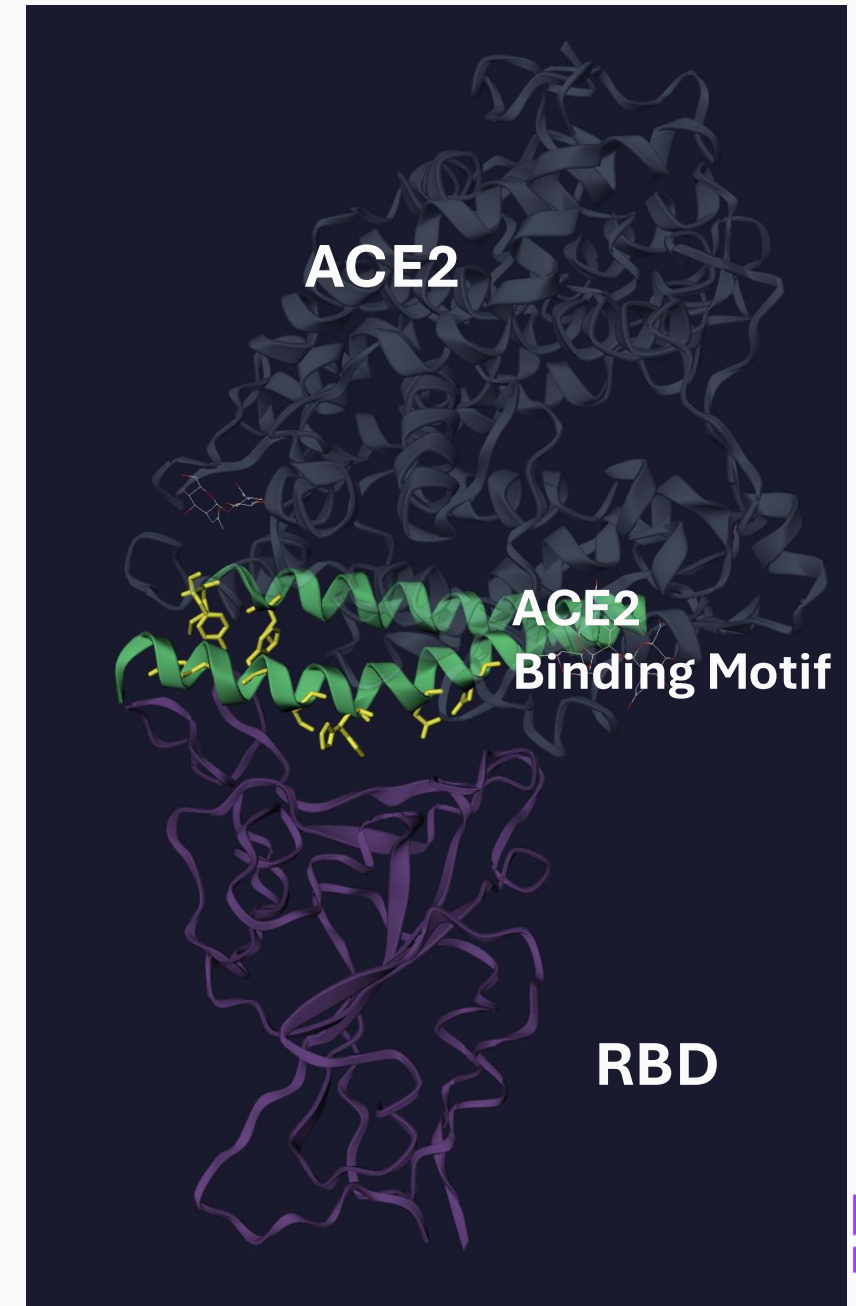
 **Protein Design Workshop 2026 - Part 2**

Dongjun Guo @ Ctrl-Alt-Bio 2026, UCL

Environment setup & Structure Visualization

Run the first three steps in the colab.
This will:

- Download Boltzgen & install the dependencies
- Check the GPU
- Enable interactive visualization of the structure



Parameter setup

We need to setup the parameters for both the binder & the target

Binder:

- What is the structural file of the binder?

8we1.cif

- Which chain (or more precisely, which residues) is the binder in the structural file?

Chain A, residues 19-83

- Which chain (or more precisely, which residues) of the binder will be considered in the design?

Here we exclude the hotspots

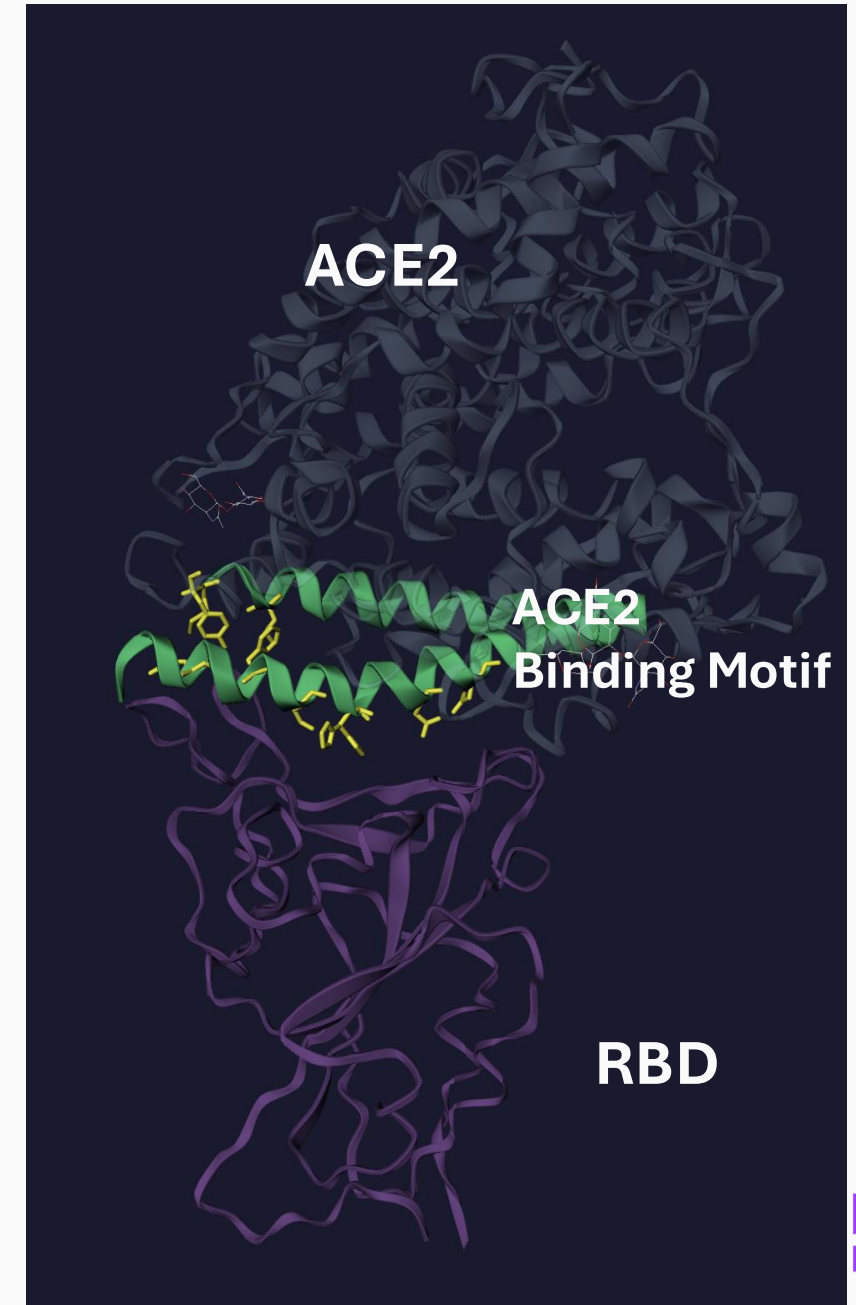
If you want to exclude 50 in fragment 1-100, write 1..49, 51..100 here.

- Do you want to also design insertions in the binder?

Yes, if you want to insert after 50, write 51 as the starting point

- Where will the binder binds to?

You can say ALL, or more specifically, give the residue index of the binding region

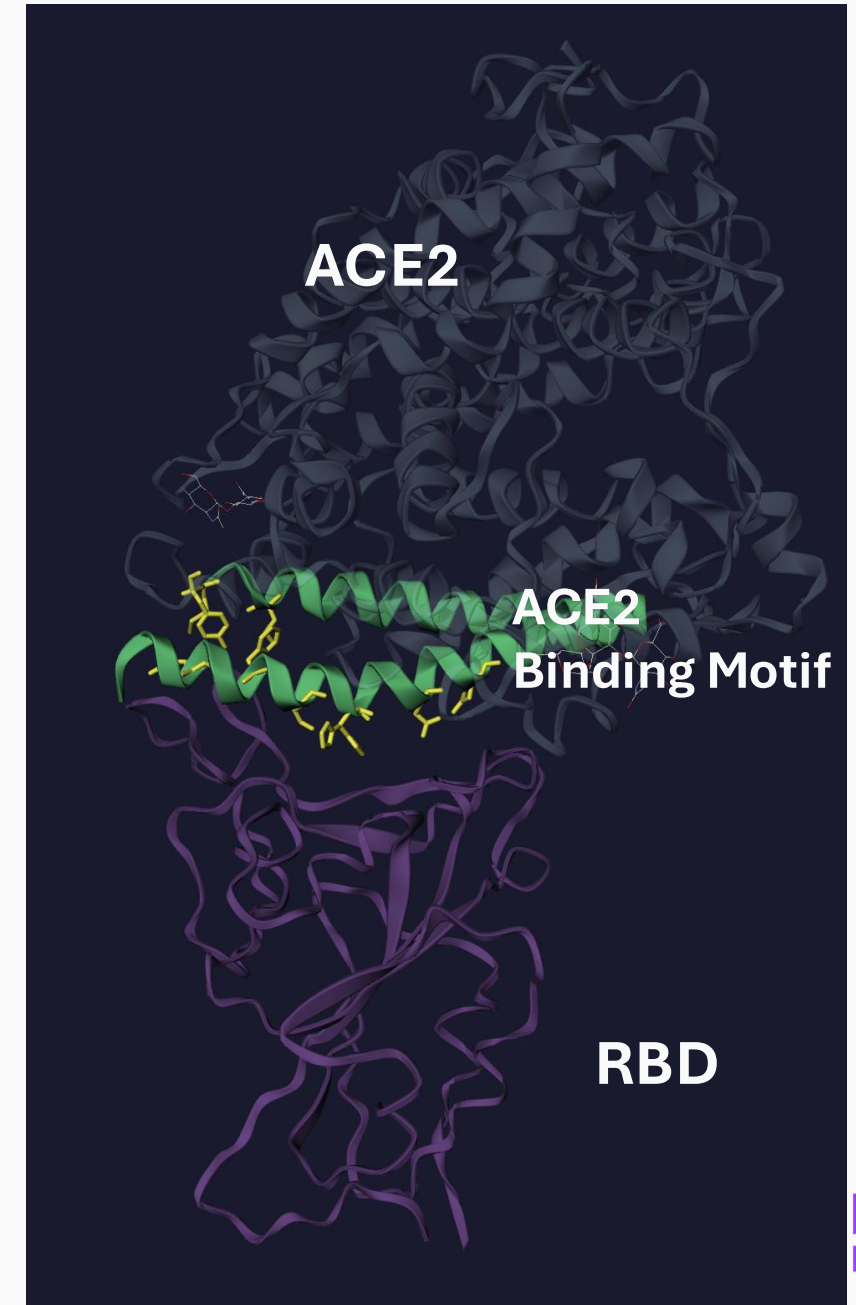


Parameter setup

We need to setup the parameters for both the binder & the target

Target:

- What is the structural file of the binder?
8we1.cif
- Which chain (or more precisely, which residues) is the binder in the structural file?
Chain B



Heads up!

- residue index in Boltzgen

Boltzgen use the `label_asym_id`, not the `auth_asym_id` author residue index

You can check the numbering in Mol* Viewer (<https://molstar.org/viewer/>)



**Ways to write
residue index:
19..82, 108**

Colab Step 4.

Input all the specified parameters, the script will automatically write out a yaml file containing all the configurations, which boltzgen will later use for the design.

```
entities:
  - file:
      path: 8we1.cif
      include:
        - chain:
            id: A
            res_index: 19..83
      design:
        - chain:
            id: A
            res_index: 19..23,25..27,29..30,32..33,35..37,39..41,43..78,80..81
      design_insertions:
        - insertion:
            id: A
            res_index: 84
            num_residues: 30..35
            secondary_structure: HELIX
      binding_types:
        - chain:
            id: B
            binding: all
  - file:
      path: 8we1.cif
      include:
        - chain:
            id: B
```

Run Boltzgen!

- Colab step 5. (~20 mins)

▶ We need to include what is the protocol to run boltzgen, how many structures we want boltzgen to generate, and how many designs we want it to preserve after the filtering step.

There are a number of different protocols ( [GitHub](#)) available in Boltzgen now. Here we use `protein-redesign` protocol.

PROTOCOL

protein-redesign ▼

NUM_DESIGNS

4

BUDGET

1

[Show code](#)

Run Boltzgen!

- Colab step 5.

Which protocols should we use?

Protocol (design-target)	Appropriate for	Major config differences
protein-anything	Design proteins to bind proteins or peptides	Includes <code>design folding</code> step.
peptide-anything	Design (cyclic) peptides or others to bind proteins	No Cys are generated in inverse folding. No <code>design folding</code> step. Don't compute largest hydrophobic patch.
protein-small_molecule	Design proteins to bind small molecules	Includes binding affinity prediction. Includes <code>design folding</code> step.
antibody-anything	Design antibody CDRs	No Cys are generated in inverse folding. No <code>design folding</code> step. Don't compute largest hydrophobic patch.
nanobody-anything	Design nanobody CDRs	Same settings as antibody-anything
protein-redesign	Redesign or optimize existing proteins	No <code>design folding</code> step. Uses <code>design_mask</code> for target/template definition.



Run Boltzgen!

- Colab step 5.

Available steps in boltzgen

Available steps:

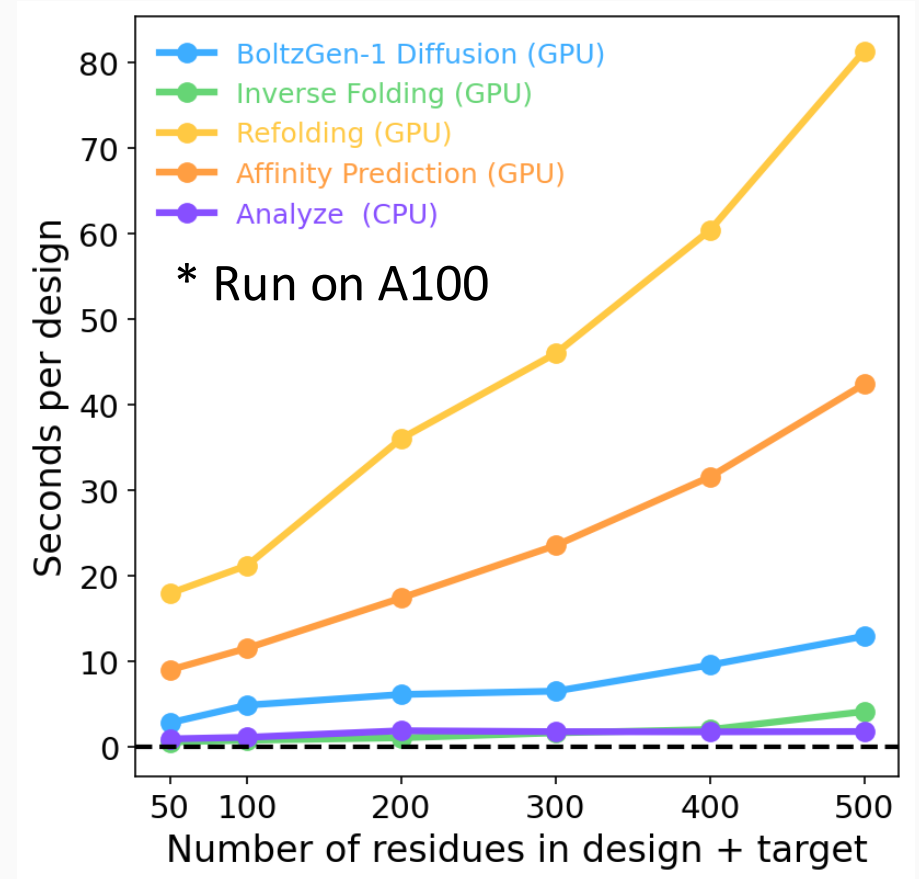
- `design` - Generate num_design candidates using the diffusion model based on your design specification
- `inverse_folding` - Redesign sequences from the previous step using our inverse folding model
- `folding` - Re-fold the designed binders with their targets using Boltz-2 model
- `design_folding` - Re-fold the designed binders alone without target (disabled for peptide and nanobody binders)
- `affinity` - Predict binding affinity between designed proteins and their target small molecules using Boltz-2 (for design of small molecule binders only)
- `analysis` - Analyze the folded structures using various metrics to assess design quality
- `filtering` - Filter and rank designs based on analysis results to select the best candidates

Run Boltzgen!

- Colab step 5.

Number: how many designs do I want boltzgen to generate?

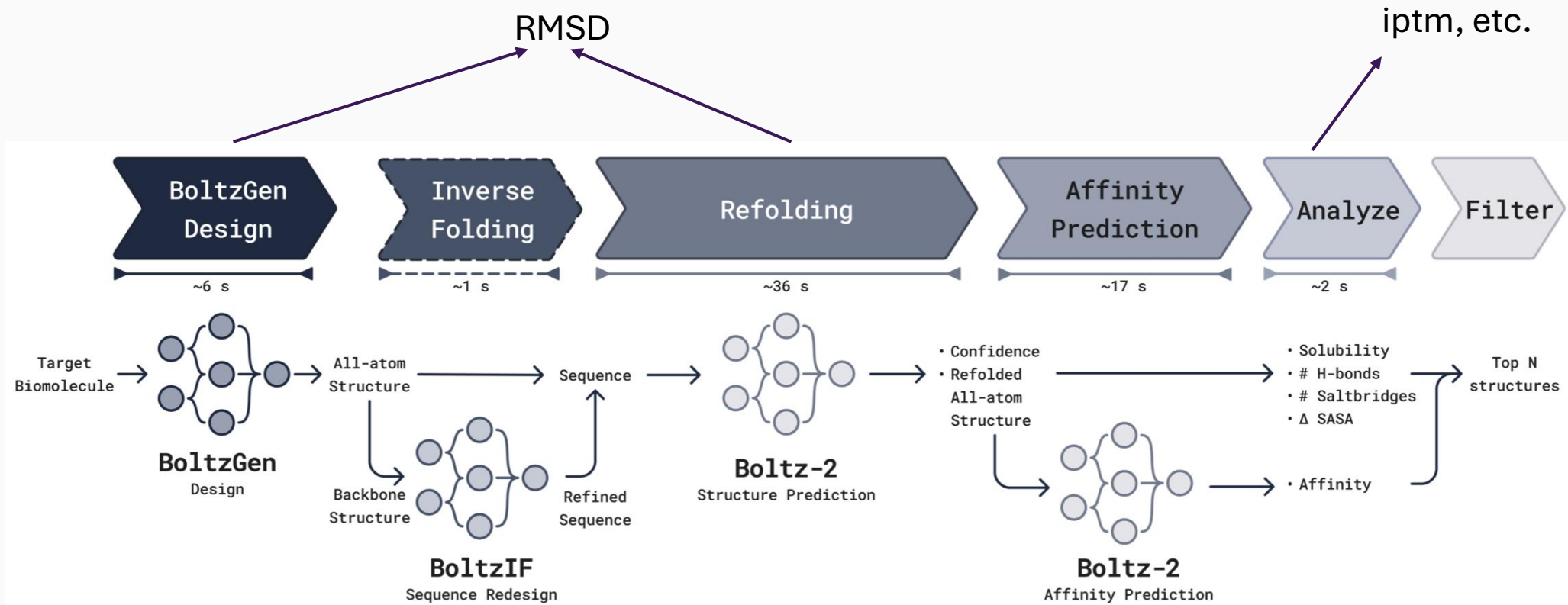
Usually run a small number on your local machine (10~50), then run a big number on HPC (10,000 – 50,000)



Run Boltzgen!

- Colab step 5.

Budget: how many designs do I want preserve after filtering step?



Run Boltzgen!

- Colab step 5.

Actual command to run boltzgen:

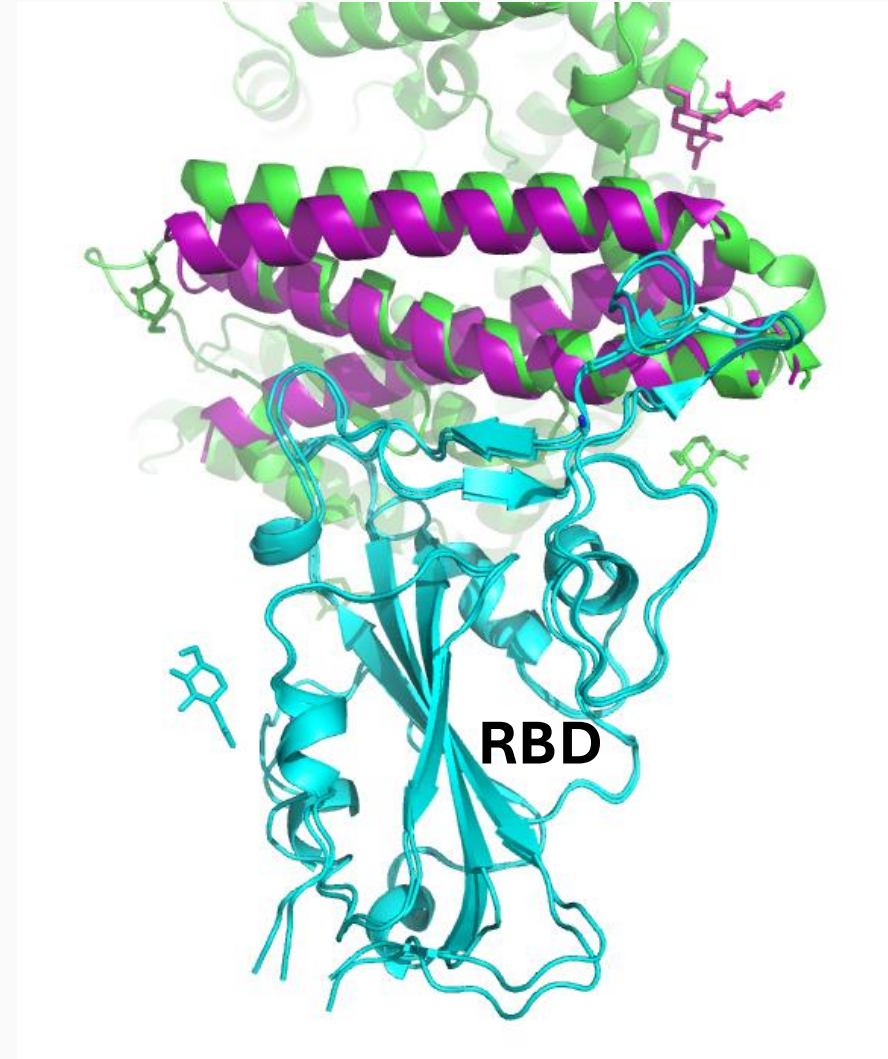
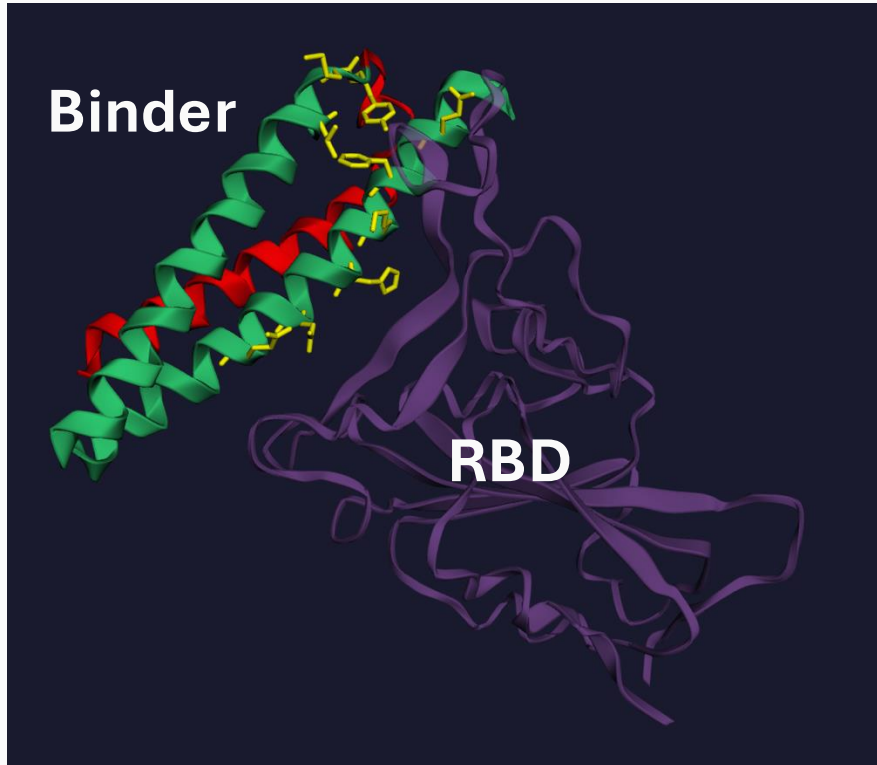
```
boltzgen run rbd_3hb_binder.yaml --output boltzgen_output --protocol protein-  
redesign --num_designs 4 --budget 1 --cache boltzgen_cache
```

Logs when you are running it:

```
Predicting: |          | 0/? [00:00<?, ?it/s]  
[Step 2/5] inverse_folding - Predicting: |          | 0/? [00:00<?, ?it/s]  
[Step 2/5] inverse_folding - Predicting: |          | 0/? [00:00<?, ?it/s]  
Predicting DataLoader 0: 0%|          | 0/4 [00:00<?, ?it/s]  
[Step 2/5] inverse_folding - Predicting DataLoader 0: : 0%|          | 0/4 [00:00<?, ?it/s]  
[Step 2/5] inverse_folding - Predicting DataLoader 0: : 100%|██████████| 4/4 [00:05<00:00, 0.80it/s]N  
  
[Step 2/5] inverse_folding - Predicting DataLoader 0: : 100%|██████████| 4/4 [00:05<00:00, 0.80it/s]  
/usr/local/lib/python3.12/dist-packages/pytorch_lightning/utilities/_pytree.py:21: `isinstance(treespe  
Initializing FromGeneratedDataModule datasets for boltzgen_output/intermediate_designs_inverse_folded  
[Info] Number of files to process (including already processed ones): 4  
Found 1 targets and 4 remaining designs that still need to be processed in this step.
```

Design Visualization

- Colab step 6.



Results analysis

- Colab step 7.

RMSD: how consistent it is between the original proposed scaffold & re-folded structure – The lower the better `filter_rmsd`

ipTM: measures the confidence of the interface, ranges from 0 to 1, the higher the better `ipTM`

Plddt: measure the confidence of the local structure of the predicted complex, the higher the better. `complex_plddt`

How boltzgen rank its design?

They will first do an independent rank for each matrix: e.g. ipTM, pTM

Then it will do an inverse weight based on the its importance (embedded in boltzgen), which will give you a final rank.

Download results & try more...

- Colab step 8

- How different it is for the binders designed by Boltzgen & RFDiffusion? – Inspect & superimpose structures of designs in pymol.
- Can we calculate the binding energy of the newly designed binder? – PBEE as shown in part1

Useful resources

In practical, parameters & protocols for each design is different. You should write your own yaml file for these configuration and run it.

Boltzgen Github provides example yaml file for most protocols, and you can adjust them accordingly. <https://github.com/HannesStark/boltzgen>



Franca Fraternali
Joseph Ng



Oriol Garcia I Carmona
Carlos Henrique Bezerra Da Cruz
Isabella (Izzy) Withnell
Tooki Chu
Cwen Li
Arianna Migliorini
Maria Laura De Sciscio



✉ d.guo@ucl.ac.uk

🐦 [@Dongjun_Guo](https://twitter.com/Dongjun_Guo)

in [Dongjun Guo](https://www.linkedin.com/in/dongjun-guo)

F **RATERNALI LAB**
@ SMB, UCL

🐦 @FraternaliLab



Thank you!